OT-AD: Optimal Transport-Guided Transformer for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) is one of the research hotspots within the domain of remote sensing image interpretation. The key is how to effectively distinguish the anomaly from the background. Existing methods usually rely on the assumption that the background of hyperspectral images (HSIs) conforms to multivariate Gaussian distributions. However, for real data, this assumption does not necessarily fit. Building upon the study of background distributions in real hyperspectral data, this article leverages optimal transport theory to model the background reconstruction task. The optimal transport of reconstruction and background is mathematically proven to be converted into that of reconstruction and the original image. To facilitate the reconstruction process, the Transformer architecture is employed due to its inherent ability to compute similarity weights. Based on the above, the optimal transport-guided Transformer anomaly detector (OT-AD) is designed by applying the Wasserstein distance constraints derived from optimal transport theory to the deep network. The experimental validation conducted on multiple public datasets confirms the effectiveness and reliability of the proposed method. The code for this work is publicly available at https://github.com/jiojio0505/OT-AD**

***Index Terms*—Background distribution, background reconstruction, hyperspectral anomaly detection (HAD), optimal transport, Transformer, Wasserstein distance constraints.**

I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data cubes that integrate both spatial and spectral information. Over the years, they have garnered increasing attention due to their exceptional resolution in the spectral dimension [1], [2], [3], [4]. The interpretation of HSIs encompasses a diverse range of tasks, such as land cover classification [5], object detection [6], and change detection [7]. Object detection tasks can be broadly classified into two categories: supervised tasks, which rely on prior knowledge of the object spectrum, and unsupervised tasks, which aim to identify unknown objects. Supervised algorithms are typically referred to as object detection, while unsupervised algorithms are commonly known as hyperspectral anomaly detection (HAD) [8]. HAD is essential for discovering potential objects of interest, with significant applications in environmental monitoring [9], disaster response [10], military reconnaissance [11], and resource exploration [12]. Anomalies differ across various scenarios. For example, in military reconnaissance, hidden troops and weapons are considered anomalies, whereas in mineral exploration, the ore of interest takes on this role.

The core of HAD is effectively separating unknown anomalies from an unknown background. According to our investigation, in the existing methods, the definition of anomaly is mainly embodied in two aspects. One aspect is that the spectral properties of anomalies differ markedly from those of the surrounding background pixels. Another is that the proportion of anomalous pixels relative to the entire image is notably small. The latter is clearly observable. Based on the characteristics of anomalies, researchers have proposed numerous approaches to achieve HAD, including traditional methods and deep learning-based methods, many of which are based on the assumption that the background follows a multivariate Gaussian distribution [13], [14], [15], [16]. To start our study with theoretical guarantees of the data distribution, we verify the above definitions and assumptions.

As shown in Fig. 1, we take three datasets as examples, synthesize their false color images, and mark the anomalous pixels as red. In addition, we take five background pixels adjacent to the anomalies and mark them in green. For ease of observation, we plot the spectral curves of the background and anomalies in the same figure. Colored curves with high transparency are anomalous spectral curves, and colored curves with low transparency are spectral curves of their neighboring background pixels. It can be seen that there are some differences between the spectral curves of anomalous pixels and their surroundings. However, for different datasets, the magnitude of these differences is not consistent. In many bands, the similarity between the two types of curves is even remarkably high.

For HSIs, a multidimensional pixel containing all spectral dimensions constitutes a multivariate vector. If the background features conform to a multivariate Gaussian distribution, then

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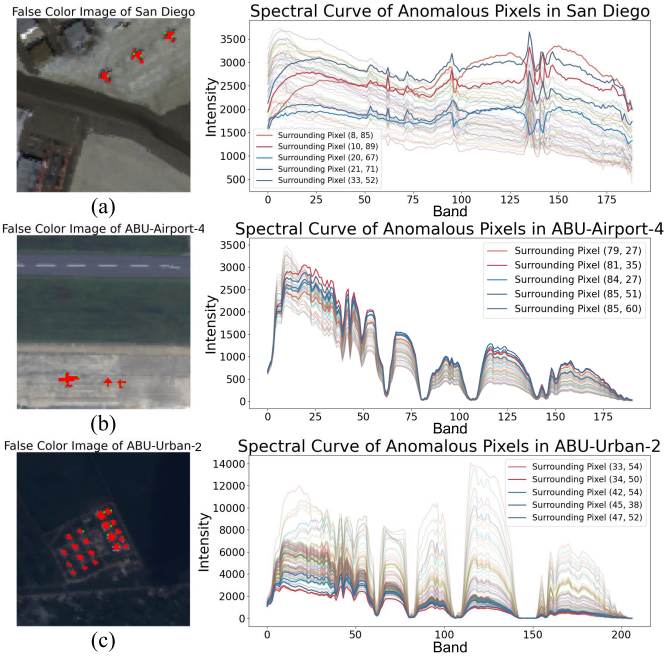


Fig. 1. Spectral curves of anomalous pixels and their adjacent pixels in HSIs. The translucent colored curves represent the spectral profiles of anomalous pixels, while the less transparent colored curves correspond to the critical background pixels. In HSIs, red pixels represent anomalous pixels, while green pixels indicate background pixels. (a) San Diego dataset. (b) ABU-Airport-4 dataset. (c) ABUUrban-2 dataset.

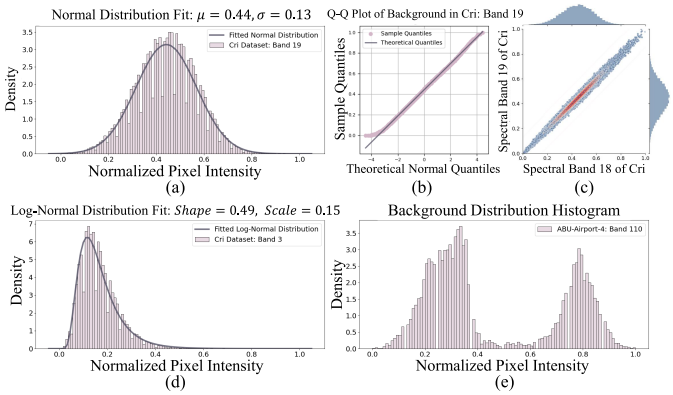


Fig. 2. Visual analysis of background distributions in HSIs. (a) Histogram of the background distribution and the fit Gaussian distribution curve for spectral band 19 of the Cri dataset. (b) Q–Q plot of the background distribution for spectral band 19 of the Cri dataset. (c) Joint distribution plot for spectral bands 18 and 19 of the Cri dataset. (d) Histogram of the background distribution combined with the fit lognormal distribution curve for spectral band 3 of the Cri dataset. (e) Histogram of the background distribution for spectral band 110 in the ABU-Airport-4 dataset.

the marginal distribution of the background in a single spectral dimension should also be Gaussian. First, we perform the Kolmogorov–Smirnov (KS) statistical test on HSIs [17], [18]. Since an HSI contains a large number of background samples in one spectral dimension, the statistical test becomes extremely strict, and, thus the Gaussian hypothesis is consistently rejected across all datasets we study. Therefore, we primarily rely on descriptive methods, such as graphical visualization, for evaluation. In particular, histograms and quantile–quantile (Q–Q) plots [19] are utilized to illustrate the background distribution across each spectral dimension. To enhance interpretability, the curves of Gaussian and lognormal distributions, fit with corresponding parameters, are superimposed onto the histograms for more intuitive observation.

We select several of the most representative visualization results from all datasets, as shown in Fig. 2. The background distribution in (a) is approximately Gaussian, (b) is its Q–Q plot, and the sample distribution is close to the theoretical distribution line. The background distribution in (d) approximately follows a lognormal pattern, while (e) exhibits a distinct bimodal feature. In (c), the joint background distribution of the band shown in (a) and its adjacent band is studied, with contour lines representing the density of data points. Overall, the data show a clear linear trend, suggesting that the background distributions of adjacent spectral bands are highly similar and share an approximate linear relationship. This offers partial support for the assumption of a multivariate Gaussian distribution. However, across all the datasets we analyze, only a few spectral bands from one dataset, Cri, yield such approximate results. In contrast, the visualization results from the other datasets do not agree with the Gaussian distribution.

To improve the generality of our method and reduce its sensitivity to variations in data distributions, we look for theories that better model the data distribution. The mathematical theory of optimal transport, focusing on optimal transportation and resource allocation, has garnered our attention. Initially proposed by French mathematician Monge [20], the theory saw significant advancement through the contributions of Soviet mathematician and economist Kantorovich [21]. In machine learning, optimal transport theory has also been applied in certain works [22]. Applying this theory, we model the background reconstruction problem for HSIs as an optimal transport problem between the distribution of the reconstruction and that of the background, and define the optimal transport cost as the Wasserstein distance. Based on the above modeling, we show that this problem can be further transformed into an optimal transport between the distribution of the reconstruction and that of the original image.

In order to verify our theory, we employ the Transformer architecture for background reconstruction, which has found wide application in computer vision for the past few years. The essence of Transformer is an encoder–decoder structure with a self-attention mechanism at its core. It was originally used in natural language processing [23] and has since been generalized to the field of computer vision [24], [25], [26]. In both the encoder and decoder, the main components are multihead self-attention (MSA) mechanisms and fully connected feed-forward networks. In them, multihead attention comprises multiple parallel scaled dot-product attention layers, enabling the model to focus on features located in different subspaces. The elements in the computed attention maps represent the similarity weights between different tokens, which are well-suited for the background reconstruction.

Leveraging Wasserstein distance constraints derived from optimal transport theory within the Transformer architecture, we propose the optimal transport-guided Transformer anomaly

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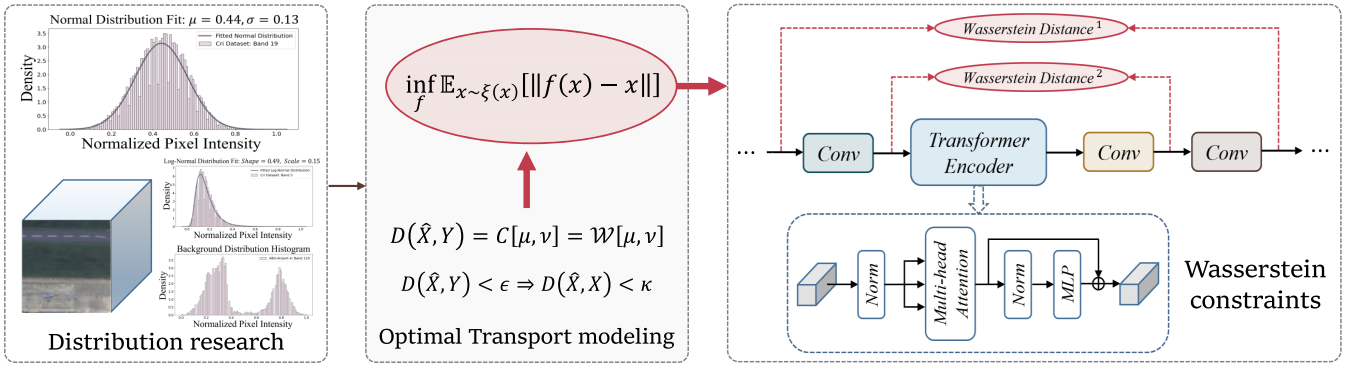


Fig. 3. Flowchart of the main contributions of this article. In particular, we first study the background and anomaly distribution of HSI, and then translate the optimization objective into the optimal transport of the reconstructed and original images according to the optimal transport theory. Finally, the optimization objective we obtained is constructed as two Wasserstein distance constraints and added to the MSA network.

detector (OT-AD). The overall contribution of this article is shown in Fig. 3, which mainly includes the following points.

1) In this article, the statistical distribution properties of HSIs are studied and analyzed. Hypothesis testing and visual verification of the applicability of the commonly used Gaussian distribution on HSIs are carried out.
2) The optimal transport theory is used to model the data distribution, and the relevant formulas and proofs are provided. In the proof, the necessity of the commonly used wavelet reconstruction is illustrated.
3) By integrating the Wasserstein distance constraints derived from optimal transport with the Transformer architecture in deep learning, we propose a novel OT-AD algorithm for HAD.

The remainder of this article is organized as follows. In Section II, we investigate the related works on HAD in recent years, including methods based on traditional research and deep learning. Section III illustrates the proposed OT-AD, including theoretical analysis and network architecture design. Section IV presents the relevant experimental results, including datasets, comparative experiments, ablation experiments, results analysis, and discussion on computational complexity. In Section V, we give the conclusion of this article.

II. Related Works

This part introduces many research methods of HAD in recent years, including methods based on traditional research and deep learning. Furthermore, we divide the methods based on deep learning into those based on autoencoder (AE), generative adversarial network (GAN), and Transformer.

A. Traditional Methods

Methods based on traditional research mainly include those based on statistical models and low-rank sparse representation. The method based on a statistical model is represented by the RX [13] algorithm. The RX algorithm is a classical HAD algorithm. It uses multivariate statistical analysis, assumes that the background characteristics obey the multivariate Gaussian distribution, and identifies outliers that do not meet the characteristics of the background as anomalies. The detection operators are obtained by calculating the mean and covariance matrix of the background. Based on the generalized likelihood ratio test, RX is a constant false alarm rate (FAR) algorithm. Many algorithms are improved based on RX, such as kernel RX [14], local RX [27], and weighted RX [28].

Based on the sparse characteristics of HSI, several HAD algorithms are proposed using sparse decomposition theory. In general speaking, in the binary classification detection model, background information usually has the characteristics of low rank, while the anomaly has the characteristics of sparsity. Representative methods include [29], [30], [31]. Although these methods are superior to traditional nonsparse anomaly detection algorithms to a certain extent, most of them still expand HSIs into 2-D planes for processing. To make the most of the 3-D characteristics of HSIs, a method based on tensor decomposition was proposed. Geng et al. [32] proposed an algorithm combining third-order statistical tensors and RX anomaly operators, which is the first literature to systematically describe the application of tensor decomposition to HAD.

In addition, there are methods based on collaborative representation [33], graph theory [34], matrix factorization [35], chessboard topology [36], differential algorithm [37], frequency domain analysis [38], and so on. These methods have been well studied and developed.

B. Deep Learning-Based Methods

With the development of deep learning, it has also been applied in HAD. Since HAD is an unsupervised task, deep generative model-based algorithms naturally become the most suitable method for it, represented by AE and GAN. Such methods mainly rely on the assumption that anomalies are rare in number and significantly different from the background, resulting in larger reconstruction errors for them. With the widespread use of vision Transformer, several works have also used it in HAD.

1) AE-Based Methods: AE performs representation learning on input data through encoding and decoding processes. It has been used to accomplish anomaly detection in HSIs by Bati et al. [39]. They sparsely encode the HSI, then decode it, and calculate the coding error as the measure of pixel anomaly. In the years that followed, there were a lot of related works.

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Zhao et al. [40] constructed a stacked denoising AE for HAD. To jointly represent the spatial–spectral features, inspired by low-rank sparse matrix factorization, they later constructed a spectral–spatial stacked AE [41], which improved the original method. Lu et al. [42] proposed an AE based on embedding manifold constraints by learning a latent representation, in order to retain the internal structure of HSIs.

Auto-AD [43] is one of the representative AE-based methods, which achieves the reconstruction of the background by a fully convolutional AE with skip connections, avoiding manual parameter setting, preprocessing, and post-processing. Xiang et al. [44] added a guided module to the AE network based on the guided image, which is calculated by the spectral similarity method, thus implementing the constraint on the anomalous features. Wang et al. [45] designed a dense residual self-attention module to extract features from the original image, and input them into AE. Xiang et al. [46] used superpixel estimation to construct a dictionary for background and anomaly, and calculated the similarity measure of each pixel point to the background and anomaly dictionary, respectively. Then, input the similarity measure into the AE network for image reconstruction. Huo et al. [47] proposed a memory-enhanced AE model, which contains a superpixel-guided adaptive weight calculation module and a sample attribution mining module, in order to improve the quality of background reconstruction while suppressing anomaly reconstruction. Sun et al. [15] embed the output of the encoder into hyperbolic space and use the wrapped Gaussian prior distribution to constrain the latent space representation. There are many other AE-based approaches for HAD [48], [49], [50], [51].

In general, the existing AE-based methods mainly include processing the original HSI as the input of the encoder, constraining the encoded latent space representation, designing the structure of the encoder and decoder, and hierarchically designing the overall architecture.

2) GAN-Based Methods: What makes GANs unique is their adversarial training mechanism. The network architecture of their generators and discriminators can be selected in a variety of ways. Using AE as the generator, a special kind of GAN called adversarial AE (AAE) is formed. Xie et al. [52], [53] first introduced the adversarial idea into HAD and proposed spectral constrained AAE. Jiang et al. [16] constructed a discriminative GAN for background reconstruction, which can suppress anomalous objects and learn an assumed Gaussian background distribution. They also add consistency-enhancing representations and shrinkage constraints in the latent feature layer, thus keeping the projection positions of anomalies similar to normal samples. Later, they adopted a weakly supervised way to learn a discriminative reconstruction model [54], using a classification method with a probability-based threshold to label coarse samples.

Jiang et al. [55] proposed a semi-supervised GAN for background modeling, which generates initial spatial features through morphological attribute filtering. Zhong et al. [56] presented a context-anomaly separability GAN, which significantly enhances the distinguishability of background and anomaly by introducing a context-anomaly separability constraint loss function. Wang et al. [57] mapped the raw spectrum of HSI to the fractional Fourier domain and proposed frequency-to-spectrum mapping GAN (FTSGAN). Introducing variational inference methods, Wang et al. [58] proposed variational background inference GAN (VBIGAN). The network consists of two sub-networks, the sample GAN and the latent GAN, to better construct the relation between samples and potential samples. In addition, a capsule differential GAN based on continuous learning was proposed to improve the cross-domain learning performance of HAD [59].

The above GAN-based methods are mainly innovated in the aspects of background distribution learning, latent space constraints, and the construction of supervision methods.

3) Transformer-Based Methods: As an important method in the field of deep learning, Transformer is used for feature representation in many tasks. It presents an encoder–decoder architecture with a self-attention mechanism as its core. In order to solve the cross-domain detection problem of HAD, Huyan et al. [60] proposed a unified deep detector AUDNet based on few-shot learning, which represents the context relation embedding by a memory Transformer model. Lian et al. [61] presented a gated Transformer network for HAD (GT-HAD) to distinguish background from anomalies using spatial–spectral similarity during HAD reconstruction. The background features and anomalous features are modeled separately by a two-branch network, and the activation states of the two branches are adjusted by an adaptive gating unit.

Aiming at the problems of low spatial resolution and poor residual detection effect in existing methods, Young et al. [62] proposed a Transformer-guided abundance domain fractional attention HAD method. The abundance matrix obtained by mix source separation of the original HSI is used as the input to the network. The core of this method is a fractional abundance attention mechanism based on a convex Q-quadratic norm. Ma et al. [63] proposed a small object perception HAD detector to capture a more robust HSI feature representation through the saliency map. In which a Transformer block with multiple fully connected layers is used for feature extraction. Wu and Wang [64] used Transformers as encoders in AE to fully exploit their global representation capabilities, thus achieving accurate modeling of high-order nonlinear mixing phenomena in HSI. He et al. [65] presented a global-to-local spatial–spectral perception Transformer network. The global sparse attention and the local self-invisible attention Transformer blocks are used to capture the multilevel deep reconstruction features.

It can be observed that methods based on Transformer primarily leverage its internal self-attention mechanism. In this way, the long-range dependencies of elements can be effectively modeled, which enables the extraction and representation of global features in HSIs.

Among the above methods, in addition to the classic RX algorithm, which uses the Gaussian distribution to model the background, there are also some recent methods that have applied the Gaussian distribution assumption. Jiang et al. [16] proposed a HAD reconstruction algorithm based on constrained GAN discriminant learning, and one of the important assumptions is that the background follows a multivariate Gaussian distribution. Yu et al. [66] proposed

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an HAD acceleration algorithm based on the multivariate Gaussian model, which determines the anomalous positions by estimating the probability distribution of the data.

III. Proposed Method

In this section, we focus on the detailed explanation of the motivation and the theoretical derivation of our method, as well as the network structure construction to verify our theory.

A. Motivation

As mentioned in Section I, we conduct statistical verification and visualization of the background distribution on several HSIs. It is found that the assumption that the background follows a multivariate Gaussian distribution is not satisfied in most cases. We would like to have a general way to model the distribution of HSIs, rather than modeling only specific data obeying certain assumptions. Our gaze turns to optimal transport, which measures the distance between probability distributions. In HAD, we want to achieve the best reconstruction of the background, which can be modeled as the optimal transport between the distribution of the reconstructed image and that of the background image. Since the background distribution in HAD is unknown, we transform the optimal transport from one between the reconstruction distribution and the background distribution to one between the reconstruction distribution and the original HSI distribution. The equivalence of these two optimal transports is proved below.

B. Equivalence of Optimal Transports

For a HSI $X \in \mathbb{R}^{H\times W\times b}$, where H denotes its height, W stands for its width, and b represents the number of its spectral bands, we break it down into two parts, background $Y \in \mathbb{R}^{H\times W\times b}$ and anomaly $A \in \mathbb{R}^{H\times W\times b}$. Like most deep learning-based methods, we proxy the HAD task with the reconstruction task of the background. The essence of the reconstruction task is to find a function f that preserves the background Y contained in X as much as possible, while suppressing the anomalies. Our goal can be formulated as follows.

1) Let N be the number of all pixels in an HSI. For $f(X) = \hat{X}$ and $\forall \epsilon > 0$, when $N \to \infty$, we pursue $D(\hat{X}, Y) < \epsilon$, where $D(\hat{X}, Y)$ is the distance between the two spaces $\hat{X}$ and Y.

Definition 1 (Kantorovich Form of Optimal Transport): Suppose μ is a probability distribution over $\hat{X}$, and ν is a probability distribution over Y, define $D(\hat{X}, Y)$ as the Kantorovich optimal transport cost between μ and ν

$$D\left(\hat{X}, Y\right) := \mathcal{C}\left[\mu, \nu\right] = \inf_{\gamma\in\Gamma(\mu,\nu)} \int_{\hat{X}\times Y} c\left(\hat{x}, y\right) d\gamma\left(\hat{x}, y\right) \tag{1}$$

where $c(\hat{x}, y) \mapsto [0, \infty)$ is the cost function, and represents the mass transported from point $\hat{x}$ to point y; $\Gamma(\mu, \nu)$ denotes the set of all probability distributions over $\hat{X} \times Y$; $\gamma(\hat{x}, y)$ represents a transport scheme from $\hat{X}$ to Y, and satisfies the edge distributions $\int \gamma(\hat{x}, y)dy = \mu(\hat{x})$ and $\int \gamma(\hat{x}, y)d\hat{x} = \nu(y)$.

For $c(\hat{x}, y)$, we choose to be the Manhattan distance raised to some power: $c(\hat{x}, y) := \|\hat{x} - y\|^{\rho}$. Thus, the Wasserstein-$\rho$ distance formula between μ and ν can be obtained

$$\begin{aligned}\mathcal{W}_{\rho}\left[\mu, \nu\right] &= \left(\inf_{\gamma\in\Gamma(\mu,\nu)} \int_{\hat{X}\times Y} \|\hat{x} - y\|^{\rho} \, d\gamma\left(\hat{x}, y\right)\right)^{\frac{1}{\rho}} \\ &= \left(\inf_{\gamma\in\Gamma(\mu,\nu)} \mathbb{E}_{(\hat{x},y)\sim\gamma(\hat{x},y)}\left[\|\hat{x} - y\|^{\rho}\right]\right)^{\frac{1}{\rho}} \\ &= \left(\mathcal{C}\left[\mu, \nu\right]\right)^{\frac{1}{\rho}}. \end{aligned} \tag{2}$$

When $\rho = 1$, the optimal transmission cost $\mathcal{C}[\mu, \nu]$ is equivalent to the Wasserstein-1 distance. For simplicity, we call $\mathcal{W}_1[\mu, \nu]$ the Wasserstein distance between μ and ν and abbreviate it to $\mathcal{W}[\mu, \nu]$. Thus, we have

$$D\left(\hat{X}, Y\right) = \mathcal{C}\left[\mu, \nu\right] = \mathcal{W}\left[\mu, \nu\right]. \tag{3}$$

However, the HAD task is unsupervised, where the distribution ν of background Y is unknown. Therefore, we introduce a probability distribution ξ over X. In Theorem 1 below, we will prove that $D(\hat{X}, Y)$ can be replaced by $D(\hat{X}, X)$.

Theorem 1: Suppose μ, ξ, and ν are probability distributions over $\hat{X}$, X, and Y, respectively, and the optimal transport cost $\mathcal{C}[\cdot, \cdot]$ is denote by $\mathcal{W}[\cdot, \cdot]$. For $\forall\epsilon > 0$, $\exists M \in \mathbb{N}$, when $N > M$, $\mathcal{C}[\mu, \nu] < \epsilon \Rightarrow \mathcal{C}[\mu, \xi] < \kappa$, and $\mathcal{C}[\mu, \xi] < \epsilon \Rightarrow \mathcal{C}[\mu, \nu] < \kappa$, where κ is a sufficiently small nonnegative number.

Proof:

$$\begin{aligned}\mathcal{C}\left[\mu, \xi\right] &= \inf_{\gamma\in\Gamma(\mu,\xi)} \int_{\hat{X}X} \|\hat{x} - x\| \, d\gamma\left(\hat{x}, x\right) \\ &= \inf_{\gamma\in\Gamma(\mu,\nu,\xi)} \int_{\hat{X}YX} \|\hat{x} - y + y - x\| \, d\gamma\left(\hat{x}, y, x\right) \\ &\le \inf_{\gamma\in\Gamma(\mu,\nu,\xi)} \int_{\hat{X}YX} \left(\|\hat{x} - y\| + \|y - x\|\right) d\gamma\left(\hat{x}, y, x\right) \\ &= \inf_{\gamma\in\Gamma(\mu,\nu)} \int_{\hat{X}Y} \|\hat{x} - y\| \, d\gamma\left(\hat{x}, y\right) \\ &\quad + \inf_{\gamma\in\Gamma(\nu,\xi)} \int_{YX} \|y - x\| \, d\gamma\left(y, x\right) \\ &= \mathcal{C}\left[\mu, \nu\right] + \mathcal{C}\left[\nu, \xi\right]. \end{aligned} \tag{4}$$

Since $\mathcal{C}[\nu, \xi]$ measures the distance between Y and X, the only difference between the two is the anomaly A. Let the number of anomalous pixels in HSI be a. Obviously, the value of a will change as N changes. The intuition is that when the scene becomes larger, the number of anomalies will also increase. So we can write a as $a(N)$.

According to the definition of anomaly, an anomaly accounts for a very small part of the whole image. Here, we reasonably assume that the growth rate of $a(N)$ with respect to N is sufficiently slow. Therefore, we have

$$\lim_{N\to\infty} \frac{a\left(N\right)}{N} = 0. \tag{5}$$

In the following, we perform a separate analysis of the term $\mathcal{C}[\nu, \xi]$ in (4)

$$\mathcal{C}\left[\nu, \xi\right] = \inf_{\gamma\in\Gamma(\nu,\xi)} \int_{YX} \|y - x\| \, d\gamma\left(y, x\right)$$

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$$= \inf_{\gamma\in\Gamma(\nu,\xi)} \int_Y \|y-x\|\, d\gamma\,(y,x) + \inf_{\gamma\in\Gamma(\nu,\xi)} \int_A \|y-x\|\, d\gamma\,(y,x). \tag{6}$$

Here, we apply a global transformation to x such that

$$x' = T(x). \tag{7}$$

T needs to meet the following two conditions.

1) On Y, x', and x are sufficiently approximate.
2) On A, there is a small upper bound δ on the difference between x' and y.

Equation (5) corresponds to the measure $\mathcal{M}(A) \to 0$ in the continuous case. In other words, the anomalous region is the zero measurement set.

By replacing x in (6) with x', we can obtain

$$\inf_{\gamma\in\Gamma(\nu,\xi)} \int_Y \left\|y-x'\right\| d\gamma\,(y,x) = \inf_{\gamma\in\Gamma(\nu,\xi)} \int_Y \left\|x-x'\right\| d\gamma\,(y,x) < \delta_1 \tag{8}$$

$$\inf_{\gamma\in\Gamma(\nu,\xi)} \int_A \left\|y-x'\right\| d\gamma\,(y,x) \le \delta\cdot\mathcal{M}(A) < \delta_2 \tag{9}$$

where δ_1 and δ_2 are sufficiently small nonnegative numbers.

For $\forall\epsilon > 0$, if $\exists M \in \mathbb{N}$, when $N > M$, $\mathcal{C}[\mu,\nu] < \epsilon$, then we have

$$\mathcal{C}\left[\nu,\xi\right] < \epsilon + (\delta_1+\delta_2) := \kappa. \tag{10}$$

Similarly,

$$\mathcal{C}\left[\mu,\nu\right] = \inf_{\gamma\in\Gamma(\mu,\nu)} \int_{\hat{X}Y} \|\hat{x}-y\|\, d\gamma\,(\hat{x},y) \le \mathcal{C}\left[\mu,\xi\right] + \mathcal{C}\left[\xi,\nu\right]. \tag{11}$$

If $\mathcal{C}[\mu,\xi] < \epsilon$, then $\mathcal{C}[\mu,\nu] < \kappa$ is easily obtained. ■

At this point, we have proved that the reconstruction target $D(\hat{X},Y) < \epsilon$ can be replaced by $D(\hat{X},X) < \kappa$. We will construct our network architecture with this goal in mind. For the transformation T in (7), we employ the wavelet transform to implement the two proposed conditions as much as possible.

C. Network Architecture

Based on our analysis of the data and theory, we do not make any statistical assumptions about the distribution of HSIs. To validate our optimal transport theory for HAD, we adopt the Transformer architecture to realize the background reconstruction, due to its inherent similarity weight computation capability.

The implementation architecture of our OT-AD can be shown in Fig. 4. In which the input image is first processed by the transformation T mentioned in Section III-B, resulting in $X = T(\text{HSI})$, which is then used as the input for the deep network. Here, we choose wavelet threshold reconstruction as T. First, the original image is decomposed by wavelet, then the high-frequency components are filtered according to the threshold, and finally, the wavelet reconstruction is carried out. This removes features that are suspected of being anomalous. For the calculation of the threshold, we have two methods: one is estimated by global statistical features, and the other is estimated by hierarchical statistical features, denoted as global and level, respectively. The global statistical feature estimation is to flatten and concatenate all the detail subbands, and calculate the overall statistics as the threshold. Hierarchical statistical feature estimation computes thresholds for different detail subbands separately.

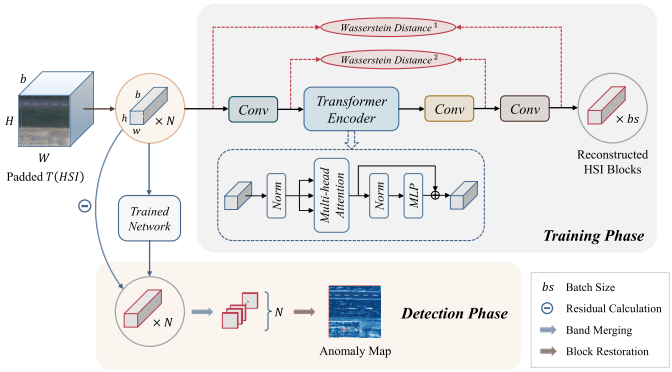


Fig. 4. Network architecture of the proposed OT-AD. It consists of a TE and three convolutional layers with two Wasserstein distance constraints.

Then, the input X is divided into N image blocks $\{X_i\}_{i=1}^N$ that overlap with each other [61]. Each block has size $h \times w \times b$ and is used as input to the reconstruction network. Such an operation can simultaneously learn information about the spatial and spectral dimensions of HSI. The reconstructed network consists of three convolutional layers and a Transformer encoder (TE). The image blocks are passed through the first convolutional layer according to the batch size, to obtain the feature map $f_{\text{conv}}^1 \in \mathbb{R}^{h\times w\times c}$. Then, f_{conv}^1 is input into TE, and after the first layer normalization (LN), MSA, the second LN, and a multilayer perceptron (MLP), the feature map after attention weight reconstruction is obtained. Two linear transformations and a nonlinear activation function are included in the MLP. The whole process of TE can be formulated as follows:

$$f_{\text{MSA}} = \text{MSA}\left(\text{LN}\left(f_{\text{conv}}^1\right)\right) \tag{12}$$

$$f_{\text{TE}} = \text{MLP}\left(\text{LN}\left(f_{\text{MSA}}\right)\right) + f_{\text{MSA}}. \tag{13}$$

The MSA is described in detail because the main background reconstruction is done in this module. Let $f_{\text{LN}} = (\text{LN}(f_{\text{conv}}^1)) \in \mathbb{R}^{h\times w\times c}$, and cut it into M nonoverlapping patches $\{P_i\}_{i=1}^M$, each P_i of size $m \times m \times c$. Apparently, we have $M = (hw)/m^2$. Then, calculate query (Q), key (K), and value (V)

$$Q = f_{\text{LN}}W^q, \quad K = f_{\text{LN}}W^k, \quad V = f_{\text{LN}}W^v. \tag{14}$$

where $W^q \in \mathbb{R}^{c\times c}$, $W^k \in \mathbb{R}^{c\times c}$, and $W^v \in \mathbb{R}^{c\times c}$ are learnable weight matrices, implemented through linear layers. In order to apply the MSA mechanism, the obtained Q, K, and V are partitioned into p portions Q_i, K_i, and V_i, according to the feature channels, where p indicates the number of self-attention heads. Next, we compute the self-attention

$$\text{Attention}_i(Q_i,K_i,V_i) = \text{softmax}\left(\frac{Q_iK_i^T}{\sqrt{d_k}}\right)V_i \tag{15}$$

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where d_k is the scale parameter, here we set it to $(\text{cm}^2)/p$. All the attention feature maps are concatenated, then transformed through a linear layer to obtain the last attention map

$$\text{Attention} = \text{Linear}\left(\text{Concat}\left(\text{Attention}_1, \ldots, \text{Attention}_p\right)\right). \tag{16}$$

The purpose of MSA is to allow the network to pay attention to the features of different spectral dimensions, and the purpose of linear transformation is to integrate the outputs of different attention heads into a unified representation. Each patch in the attention feature map obtained by (16) is reconstructed from itself and all the remaining patches according to the attention weights. For anomalous pixels, the two dilutions of patch division and surrounding patch reconstruction can effectively suppress their reconstruction.

After f_{TE} obtained by TE passes through two convolutional layers successively, the final reconstructed HSI block $\hat{X}_i$ is obtained, the loss is calculated in batches according to the batch size, and backpropagated. In Section III-B, we convert $D(\hat{X}, Y)$ to $D(\hat{X}, X)$, and give the proof. Finally, the goal of our optimization becomes

$$\begin{aligned}\mathcal{W}\left[\mu, \xi\right] &= \inf_{\gamma \in \Gamma(\mu,\xi)} \mathbb{E}_{(\hat{x},x)\sim\gamma(\hat{x},x)}\left[\|\hat{x} - x\|\right] \\ &= \inf_{f} \mathbb{E}_{x\sim\xi(x)}\left[\|f(x) - x\|\right].\end{aligned} \tag{17}$$

Based on (17), we design the loss of OT-AD, which is composed of the commonly used $L1$ loss and two Wasserstein distance constraints

$$\begin{aligned}\text{Loss} = \|f(x) - x\|_1 &+ \alpha_1 \cdot \mathbb{E}_{x\sim\xi(x)}\left[\|f(x) - x\|\right] \\ &+ \alpha_2 \cdot \mathbb{E}_{x\sim\xi(x)}\left[\left\|f'(x) - x'\right\|\right]\end{aligned} \tag{18}$$

where $f(x) = \hat{x}$, $x' = f^1_{\text{conv}}$, $f'(x) = (f_{\text{TE}})^2_{\text{conv}}$. α_1 and α_2 are hyperparameters that control the weights of the two Wasserstein distance constraints.

The physical significance of the two Wasserstein distance constraints we design can be explained as follows.

1) *Wasserstein Distance^1:* Computes the Wasserstein distance between the input image x and the reconstruction result $\hat{x}$, which represents the statistical-level consistency of the signal. This ensures that the reconstructed image aligns with the original signal in terms of pixel-level and distributional characteristics, thereby guaranteeing the authenticity and completeness of the results.
2) *Wasserstein Distance^2:* Calculates the Wasserstein distance between feature maps f^1_{conv} and $(f_{\text{TE}})^2_{\text{conv}}$. Since the reconstruction operation is implemented within the TE, both the feature maps before and after the module can be broadly regarded as "input" and "output." In order to align the global semantic representation output f_{TE} by the TE with f^1_{conv} obtained through convolution processing at the same scale, we design the second constraint between f^1_{conv} and $(f_{\text{TE}})^2_{\text{conv}}$. This constraint maintains the stability of the signal structure in the deep feature space, ensuring the continuous transmission of abstract features across different levels. It prevents information deviation and helps ensure the structural and semantic consistency of the reconstruction results.

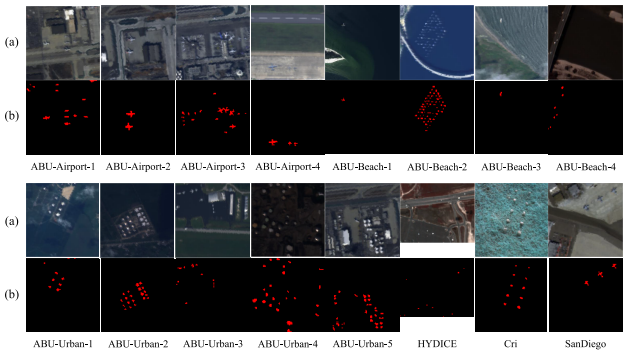


Fig. 5. False color images and ground-truth images of anomalies for different datasets. (a) False color images. (b) Ground truth of anomalies.

In the detection phase of anomalies, we use the above trained network for all HSI blocks $\{X_i\}_{i=1}^N$, to obtain their reconstructed images $\{\hat{X}_i\}_{i=1}^N$. At this point, we can compute the residual between X_i and $\hat{X}_i$: $R_i = |X_i - \hat{X}_i| \in \mathbb{R}^{h\times w\times b}$. We then apply the HSI block generation in the reverse way, restoring $\{R_i\}_{i=1}^N$ to the original HSI size, and averaging the overlaps to get the residual cube $R \in \mathbb{R}^{H\times W\times b}$. The spectral dimensions of R are finally summed to obtain the anomaly detection map R_{final}.

IV. Experimental Results

In this section, we conduct relevant experimental verification of our OT-AD. First, 16 public datasets used in the experiments are introduced. Then, the selected nine comparison methods, experimental environment, and evaluation metrics are introduced. Next, the experimental results of different methods on different datasets are visualized and analyzed. To investigate the effect of the Wasserstein distance between features at different locations of the network, ablation experiments are performed. Finally, the algorithm complexity is analyzed based on the running time of different algorithms.

A. Datasets

We conduct experiments on 16 publicly available datasets containing images in multiple scenarios acquired by multiple sensors, such as AVIRIS, ROSIS, and HYDICE. We summarize the basic information of all datasets in Table I. Besides, Fig. 5 displays the false color images and ground truth of anomalies from all datasets.

B. Experimental Setup

This section describes our experimental setup, including the selected comparison methods, all the experimental environments, implementation details, and the evaluation metrics adopted in the experiments.

1) Comparison Methods: We selected five traditional methods and four deep learning-based methods for comparative experiments with our OT-AD. The traditional methods include the classical methods RX [13], CRD [33], GTVLRR [67], PTA [68], and the new method IEEPST [69]. Deep learning-based methods include the classical method Auto-AD [43] and the new methods DirectNet [70], GT-HAD [61], and DFAN [50].

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TABLE I

INTRODUCTION TO THE 16 PUBLICLY AVAILABLE DATASETS USED IN THE EXPERIMENTS

Parameters	Airport-1	Airport-2	Airport-3	Airport-4	Beach-1	Beach-2	Beach-3	Beach-4	Urban-1	Urban-2	Urban-3	Urban-4	Urban-5	HYDICE	Cri	SanDiego
Sensor	AVIRIS	AVIRIS	AVIRIS	AVIRIS	AVIRIS	AVIRIS	AVIRIS	ROSIS	AVIRIS	AVIRIS	AVIRIS	AVIRIS	AVIRIS	HYDICE	Nuance Cri	AVIRIS
Region	Los Angeles	Los Angeles	Los Angeles	Gulfport	Cat Island	San Diego	Bay Champagne	Pavia	Texas Coast	Texas Coast	Gainesville	Los Angeles	Los Angeles	California	Grassy	San Diego
Resolution (m/pixel)	7.1	7.1	7.1	3.4	17.2	7.5	4.4	1.3	17.2	17.2	3.5	7.1	7.1	2.0	–	3.5
Spatial Size	100×100	100×100	100×100	100×100	150×150	100×100	100×100	150×150	100×100	100×100	100×100	100×100	100×100	80×100	400×400	100×100
Bands	205	205	205	191	188	193	188	102	204	207	101	205	205	162	46	189
Anomalies	144	87	170	60	19	202	11	68	67	155	52	272	232	17	1254	58

[1] We omit the ABU- prefix here due to space limitations.

TABLE II

PARAMETER SETTINGS OF THE COMPARISON METHOD IEEPST

Parameters	Airport-1	Airport-2	Airport-3	Airport-4	Beach-1	Beach-2	Beach-3	Beach-4	Urban-1	Urban-2	Urban-3	Urban-4	Urban-5	HYDICE	Cri	SanDiego
N_{topo}	60	30	100	140	150	100	20	60	70	20	30	30	50	140	180	40
per_{ie}	0.7	0.8	0.9	0.6	0.3	1.0	0.9	0.4	0.9	0.9	1.0	1.0	0.7	0.6	0.1	0.9

[1] We omit the ABU- prefix here due to space limitations.

TABLE III

PARAMETER SETTINGS OF THE COMPARISON METHOD DFAN

Parameters	Airport-1	Airport-2	Airport-3	Airport-4	Beach-1	Beach-2	Beach-3	Beach-4	Urban-1	Urban-2	Urban-3	Urban-4	Urban-5	HYDICE	Cri	SanDiego
$MinPts$	3	6	6	2	3	2	3	6	4	3	4	6	2	3	6	3
Eps	0.5	0.5	0.5	0.5	0.5	0.5	0.3	0.6	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
$Epochs$	30	20	20	20	30	20	10	20	20	30	20	20	20	30	20	30
LR	10^{-2}	10^{-2}	10^{-2}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-2}	10^{-4}	10^{-2}	10^{-4}	10^{-2}	10^{-2}	10^{-2}

[1] We omit the ABU- prefix here due to space limitations.

We reimplemented all the compared methods, ensuring consistency in data normalization, computational resources and environment, software configurations, and performance metric calculations. For setting the experimental parameters, we have the following considerations. The RX algorithm is a classical anomaly detection algorithm based on a statistical model. The detection operator is derived from the mean and covariance matrix of the image, eliminating the need for parameter settings. CRD is an anomaly detection algorithm based on collaborative representation that incorporates a distance-weighted regularization matrix into the optimization problem, using a positive regularization parameter λ. In addition, CRD employs a two-window strategy and, thus has parameters for inner and outer window sizes w_{in} and w_{out}. After experiments, we adopt parameter settings $\lambda = 10^{-6}$, $w_{\text{in}} = 15$, and $w_{\text{out}} = 19$ that perform well on all datasets.

GTVLRR is a low-rank representation model that incorporates both graph regularization and total variation regularization. The objective function of the method includes the parameters λ, β, and γ of the equilibrium. There are also the number of nearest neighbors k and the scalar parameter σ. We follow the original paper and set these parameters to $\lambda = 0.5$, $\beta = 0.2$, $\gamma = 0.05$, $k = 10$, and $\sigma = 1$. PTA decomposes HSI into background tensors and anomaly tensors, and uses different norms to regularize the prior of different dimensions. It has five parameters: the truncated low-rank r, and the hyperparameters α, μ, β, and τ. Following the setting of the original paper, we have $\alpha = 1$ and $\tau = 1$. In addition, we set $\beta = 1$, which achieves better results than $\beta = 0.01$ in the original paper. According to the authors, most datasets have r in the range 0–20. To reduce the training cost, we choose the median $r = 10$. For μ, we choose the optimal value in the set $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4\}$ for each dataset. As a result, in the order of the 16 datasets, we set $\{10^{-2}, 10^{-2}, 10^{-4}, 10^{-2}, 10^{-3}, 10^{-3}, 10^{-3}, 10^{-4}, 10^{-1}, 10^{-4}, 10^{-3}, 10^{-4}, 10^{-3}, 10^{-3}, 10^{-4}, 10^{-2}\}$. IEEPST is a method that combines point set topology and information entropy theory. It has two parameters: the number of point sets N_{topo} in each topological space, and the percentage per_{ie} used to calculate the threshold to constrain the information entropy. Referring to the parameter settings in the original paper and our experimental verification, we set the optimal parameter setting scheme as shown in Table II for different datasets.

Auto-AD is an autonomous, fully convolutional AE detection network without manually setting parameters. DirectNet integrates the sliding dual-window model into the blind block network, with parameters of the dual-window sizes W_{in} and W_{out}, and the number of ResNet blocks N_r. Among which, W_{in} is fixed to 3 and $N_r = (W_{\text{out}} - 7)/4$, so only the value of W_{out} needs to be adjusted. Due to the long training time of this algorithm, we choose the modes of the parameters on different datasets in the original paper to set, specifically, $W_{\text{out}} = 19$ and epochs = 700. GT-HAD introduces an adaptive gating unit to model the background and anomalous features without manually setting parameters. DFAN uses an adaptive aggregation model to divide the background into multiple modes. The authors discuss the effect of four parameters on the performance of the experiment, which are MinPts for counting background categories, Eps for counting anomaly categories, Epochs of the experiment, and learning rate (LR). Following the settings in the original paper, we set the parameters for different datasets in turn as Table III.

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TABLE IV
DEFAULT PARAMETER SETTINGS FOR OT-AD

Parameters	Meanings	Values
h	Height of HSI block	15
w	Width of HSI block	15
c	Number of feature channels	64
m	Size of image patch	3
p	Number of self-attention heads	2
α_1	Weight of Wasserstein constraint 1	1
α_2	Weight of Wasserstein constraint 2	1
$epoch$	Number of training rounds	150
lr	Learning rate	10^{-3}
t	Wavelet threshold generation mode	Global

2) Experimental Environments: All the deep learning-based methods, including Auto-AD, DirectNet, GT-HAD, DFAN, and our OT-AD, are experimentally performed on NVIDIA RTX 3090 (24 GB) * 1 GPU. The deep learning framework we use is PyTorch 2.1.0, the computing platform is Cuda 12.1, and the version of Python is Python 3.10 (Ubuntu 22.04).

All traditional methods, including RX, CRD, GTVLRR, PTA, and IEEPST, are tested on the MATLAB R2021b platform, which is installed on an Intel[1] Core[2] i7-9700 CPU at a 3.00-GHz computer with 16.0-GB RAM.

3) Implementation Details: We adopt the adaptive moment estimation (Adam) algorithm as the optimizer of the network. Besides, there are some default parameters that need to be set during the training of our OT-AD, as shown in Table IV.

In the experiment, we adjust the four parameters t, p, α_1, and α_2. For different datasets, we get the following optimal parameter settings in Table V.

4) Evaluation Metrics: We compare the detection results of all methods through three aspects of evaluation metrics. Subjectively, the anomaly detection result map is used to judge the detection of anomalies, which is a qualitative metric. The results are objectively evaluated using receiver operating characteristic (ROC) curves and the area under the curve (AUC), both of which are quantitative metrics. The x-axis of the ROC curves represents the FAR, while the y-axis represents the probability of detection (PD). PD is the ratio of the number of pixels correctly detected as belonging to an anomaly over all the true anomaly pixels, and FAR refers to the proportion of the background pixels that are mistaken as an anomaly over all the background pixels. The closer the ROC curve is to the top left corner, the better the model's performance. The AUC value, representing the area under the ROC curve, indicates better performance when it is closer to 1. By incorporating the threshold parameter, we also generate 3-D ROC curves, allowing for a more comprehensive analysis of the results.

In addition, box-and-whisker plots [71] are usually used to evaluate the separability of the algorithm, that is, the algorithm's capability to suppress the background and distinguish the anomaly. Two separability maps, the anomalous box and background box, can be obtained by statistically calculating the normalized detection values of anomalous pixels and background pixels. Each box contains data between the two quartiles. The line in the center of the box represents the median of the data, reflecting its average level. The top of the whisker indicates the maximum value of the data, while the bottom represents the minimum value. The lower the background box, the more severely the background is suppressed. The distance between the background box and the anomaly box indicates the algorithm's separation of background and anomaly, and the larger the distance, the better the separation of the method.

[1]Registered trademark.
[2]Trademarked.

C. Performance Comparison

In this section, we use the evaluation metrics described above to conduct a comprehensive analysis of the results obtained by our selected comparison methods.

We selected the results of several well-performing datasets to present the anomaly detection heatmap, as shown in Fig. 6. All anomaly maps are single-shot result maps close to the average AUC values in Table VI. We can see that our OT-AD achieves good background suppression and anomaly response on ABU-Airport-3, ABU-Beach-4, ABU-Urban-2, and ABU-Urban-4. For ABU-Airport-4 and ABU-Urban-3, although the background is not perfectly suppressed, the degree of anomalous response is quite prominent. Overall, our method successfully balances background suppression and anomaly response. For the comparison methods, we can see that RX, CRD, IEEPST, Auto-AD, and DirectNet do a good job in background suppression, but they also suppress anomalies to a certain extent, which is risky in practical applications. For example, for the three aircraft anomalies in ABU-Airport-4, IEEPST only weakly responds to the aircraft on the left, and basically fails to identify the two small aircraft on the right. Similarly, IEEPST does not respond as expected for many anomalous pixels in ABU-Urban-3 and ABU-Urban-4. There is more noise in the results of GTVLRR and DFAN, which indicates that their background suppression is insufficient, which will lead to a higher FAR. The results of PTA are not particularly visually unreasonable, but there are still some false detections and missed detections, such as a large anomalous response to the right background pixels of ABU-Urban-3, and a lack of due response to many anomalies in ABU-Urban-4. The anomaly maps of GT-HAD are very beautiful, and the average pooling post-processing operation in GT-HAD can diffuse the anomalous response. However, it also diffuses the response of the misdetected background, which affects the detection results.

The ROC curves corresponding to the above anomaly detection maps are shown in Fig. 7. The dark red bold curves indicate the results of our OT-AD. The 2-D and 3-D ROC curves of our method are clearly positioned in the upper left corner for ABU-Urban-2, ABU-Urban-3, and ABU-Urban-4. For ABU-Airport-3, ABU-Airport-4, and ABU-Beach-4, although the middle section of the curve is partially below the other methods, it is still in the upper left corner as a whole. Table VI presents the AUC values for all methods. In order to enhance confidence in the results, we take the

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TABLE V
OPTIMAL PARAMETER SETTINGS OF OT-AD ON DIFFERENT DATASETS

Parameters	Airport-1	Airport-2	Airport-3	Airport-4	Beach-1	Beach-2	Beach-3	Beach-4	Urban-1	Urban-2	Urban-3	Urban-4	Urban-5	HYDICE	Cri	SanDiego
t	Level	Level	Global	Level	Level	Global	Global	Level	Global	Level	Global	Level	Global	Level	Global	Level
p	2	4	2	4	2	4	4	2	2	2	4	4	2	2	2	2
α_1	0.4	0.8	0.6	0.4	0.0	0.0	0.4	0.2	0.1	0.8	0.4	0.5	0.6	0.9	0.4	0.5
α_2	0.1	0.2	0.5	0.3	0.8	0.1	0.0	0.8	0.7	0.6	0.0	0.7	0.3	0.7	1.0	0.5

[1] We omit the ABU- prefix here due to space limitations.

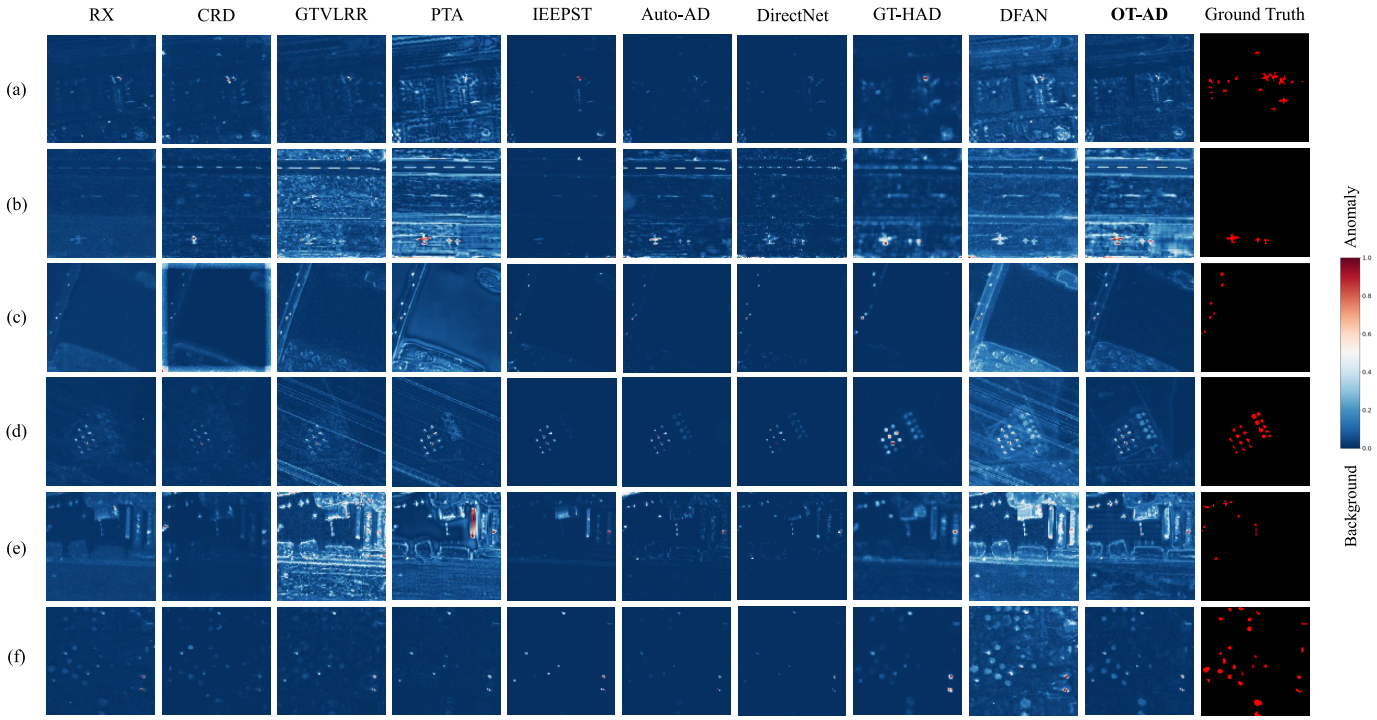


Fig. 6. Heatmaps of anomaly detection results of different HAD methods on six publicly available datasets. (a) ABU-Airport-3. (b) ABU-Airport-4. (c) ABU-Beach-4. (d) ABU-Urban-2. (e) ABU-Urban-3. (f) ABU-Urban-4.

average AUC obtained from five runs of all methods as their final AUC value, except for DirectNet. Bold numbers, underlined numbers, and italic numbers indicate the highest, second-highest, and third-highest scores, respectively. It can be seen that our OT-AD achieves the highest scores on all six datasets, which is consistent with the ROC curves. It is worth noting that on all 16 datasets, our method achieves the best results on seven datasets and ranks in the top three on 13 datasets. Moreover, the average AUC reaches 0.9783, which was 0.39% higher than the second place. This is a very impressive achievement.

Fig. 8 displays the box-and-whisker plots for various methods across the experimental datasets. In which the blue box shows the distribution of the anomaly, and the pink box shows the distribution of the background. The white line in the center of the box indicates the median, while the top and bottom edges represent the two interquartile points. The ends of the whisker denote the maximum and minimum values of the data, respectively. The greater the distance between the blue and pink boxes, the better the algorithm's separation of anomalies and backgrounds. It is observed that our OT-AD gets the best separability results on all six example datasets. The gap between the background box and the anomalous box is substantial, with almost no overlap in the data distributions of the two.

D. Ablation Study

We now conduct ablation experiments targeting the way the Wasserstein distance constraint is added. In particular, we discuss the following three cases respectively: 1) no Wasserstein distance constraint; 2) only Wasserstein constraint 1 exists; 3) only Wasserstein constraint 2 exists; 4) the two Wasserstein constraints exist simultaneously. Note that in each setting, instead of simply setting the weights α_1 or α_2 to 0, we completely remove the other parts, in order to more clearly discern the contribution of different constraints to model performance. Moreover, we set all the parameters to the default values in Table IV.

We randomly selected six datasets as examples. In Table VII shows the AUC results of our ablation experiments. We can see that the performance of the network before and after adding our Wasserstein constraints is significantly improved, especially on ABU-Beach-3, ABU-Urban-3, and HYDICE; their AUC values are increased

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TABLE VI
AUC SCORES OF TEN HAD METHODS ON 16 PUBLICLY AVAILABLE DATASETS

Datasets	RX	CRD	GTVLRR	PTA	IEEPST	Auto-AD	DirectNet	GT-HAD	DFAN	OT-AD
ABU-Airport-1	0.8221	0.8459	0.8987	0.8809	0.7686	0.9080	0.8871	**0.9644**	*0.9396*	0.9554
ABU-Airport-2	0.8404	0.9292	0.8712	0.9085	0.8641	0.8876	0.8985	**0.9858**	0.9647	*0.9519*
ABU-Airport-3	0.9288	0.8725	0.9273	0.9408	0.8499	0.8703	0.8897	*0.9482*	0.9542	**0.9566**
ABU-Airport-4	0.9526	0.9357	0.9812	*0.9946*	0.9572	0.9730	0.9541	0.9957	0.9915	**0.9966**
ABU-Beach-1	0.9807	0.9947	0.9738	0.9827	0.9876	0.9711	0.9787	**0.9996**	*0.9925*	0.9863
ABU-Beach-2	0.9106	0.8243	0.9300	0.9154	0.8968	0.8819	*0.9299*	0.9254	0.9236	**0.9302**
ABU-Beach-3	**0.9999**	**0.9999**	0.9918	*0.9993*	0.9651	0.9987	0.9961	0.9997	0.9990	0.9971
ABU-Beach-4	0.9538	0.8348	0.9797	0.9658	0.9509	*0.9824*	0.9802	0.9851	0.9779	**0.9875**
ABU-Urban-1	0.9907	0.8864	0.8272	0.9748	0.9890	0.9915	0.9829	**0.9976**	*0.9921*	0.9926
ABU-Urban-2	0.9946	0.9634	0.8596	0.9867	0.9988	0.9938	0.9877	*0.9952*	0.9903	**0.9993**
ABU-Urban-3	0.9513	0.9154	0.9224	*0.9704*	0.9495	0.9679	0.9703	0.9710	0.9693	**0.9925**
ABU-Urban-4	*0.9887*	0.9426	0.9169	0.8992	0.9907	0.9881	0.9645	0.9884	0.9759	**0.9968**
ABU-Urban-5	*0.9692*	0.9100	0.9259	**0.9820**	0.9607	0.8867	0.9481	0.9447	0.9647	0.9714
HYDICE	0.9857	0.9415	0.9908	0.9219	0.9235	**0.9975**	0.9884	0.9035	0.9951	*0.9909*
Cri	0.9675	0.9784	0.5488	0.8328	0.9951	0.5149	0.6240	0.9688	**0.9953**	*0.9830*
SanDiego	0.8885	0.9644	0.9723	**0.9869**	0.9341	0.8606	0.9589	0.7607	*0.9652*	0.9642
Average	0.9453	0.9212	0.9074	0.9464	0.9364	0.9171	0.9337	*0.9584*	0.9744	**0.9783**

[1] Bold numbers indicate the highest AUC score, underlined numbers indicate the second highest AUC score, and italic numbers indicate the third highest AUC score.

[2] All the results in the table are the average results obtained by these algorithms over five runs, except for DirectNet due to its long training time.

TABLE VII
AUC SCORES ON SIX DATASETS IN THE ABLATION STUDY OF OT-AD

Constraint Setting	ABU-Airport-4	ABU-Beach-3	ABU-Urban-2	ABU-Urban-3	ABU-Urban-4	HYDICE	Average
WD^{no}	0.9787	0.9086	0.9847	0.8874	0.9920	0.7779	0.9216
Time (s)	3.86	3.36	3.65	3.47	3.66	2.59	3.43
WD^1	0.9926 (+ 1.39%)	**0.9941 (+ 8.55%)**	0.9988 (+ 1.41%)	0.9876 (+ 10.02%)	**0.9950 (+ 0.30%)**	**0.9921 (+ 21.42%)**	**0.9934 (+ 7.18%)**
Time (s)	5.50 (+ 1.64)	4.88 (+ 1.52)	5.12 (+ 1.47)	4.93 (+ 1.46)	5.17 (+ 1.51)	3.45 (+ 0.86)	4.84 (+ 1.41)
WD^2	**0.9952 (+ 1.65%)**	0.9939 (+ 8.53%)	0.9988 (+ 1.41%)	0.9885 (+ 10.11%)	0.9814 (- 1.06%)	0.9914 (+ 21.35%)	0.9915 (+ 6.99%)
Time (s)	5.56 (+ 1.70)	4.90 (+ 1.54)	5.06 (+ 1.41)	4.96 (+ 1.49)	5.04 (+ 1.38)	3.49 (+ 0.90)	4.84 (+ 1.41)
WD^{12}	0.9915 (+ 1.28%)	0.9931 (+ 8.45%)	**0.9990 (+ 1.43%)**	**0.9899 (+ 10.25%)**	0.9943 (+ 0.23%)	0.9912 (+ 21.33%)	0.9932 (+ 7.16%)
Time (s)	6.05 (+ 2.19)	5.56 (+ 2.20)	5.86 (+ 2.21)	5.68 (+ 2.21)	5.87 (+ 2.21)	3.91 (+ 1.32)	5.49 (+ 2.06)

[1] Bold numbers indicate the highest AUC score on one dataset with different Wasserstein constraints.

[2] *WD* indicates the Wasserstein distance constraint, and WD^{no} indicates that there is no Wasserstein constraints.

[3] All the results in the table are the average results obtained by OT-AD over five runs.

by 8.55%, 10.25%, and 21.42%, respectively. There is no necessary connection between the different constraints. For example, on ABU-Airport-4, WD^2 achieves the best score; on ABU-Beach-3, ABU-Urban-4 and HYDICE, WD^1 achieves the highest score; on ABU-Urban-2 and ABU-Urban-3, WD^{12} win the first place. For ABU-Urban-4, because the Transformer architecture itself gets a good background reconstruction effect, the AUC value decreases after adding the WD^2 constraint, but the results under the other two constraints are improved. In addition, we also calculate the running times of the algorithm under different Wasserstein constraints. It can be seen that increasing a single Wasserstein constraint increases the running time, and different constraints are independent of each other. The longest running times occur when both constraints exist. However, since the construction of our model conforms to the theoretical analysis results and the distribution of the data itself, and there are no unnecessary branches in the network to consume the running time, the overall efficiency of our algorithm is high.

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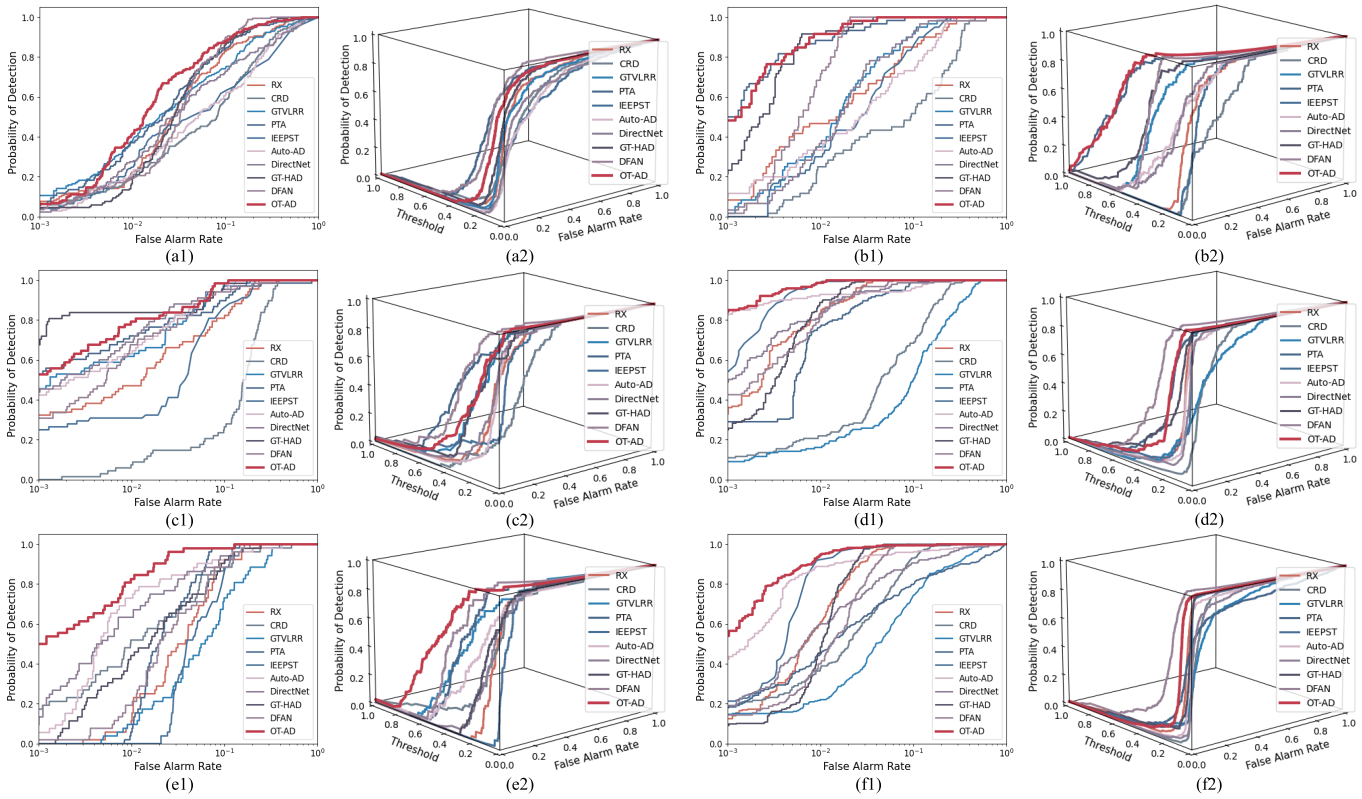


Fig. 7. 2-D and 3-D ROC curves for various HAD methods across six publicly available datasets. (a1) and (a2) ABU-Airport-3. (b1) and (b2) ABU-Airport-4. (c1) and (c2) ABU-Beach-4. (d1) and (d2) ABU-Urban-2. (e1) and (e2) ABU-Urban-3. (f1) and (f2) ABU-Urban-4.

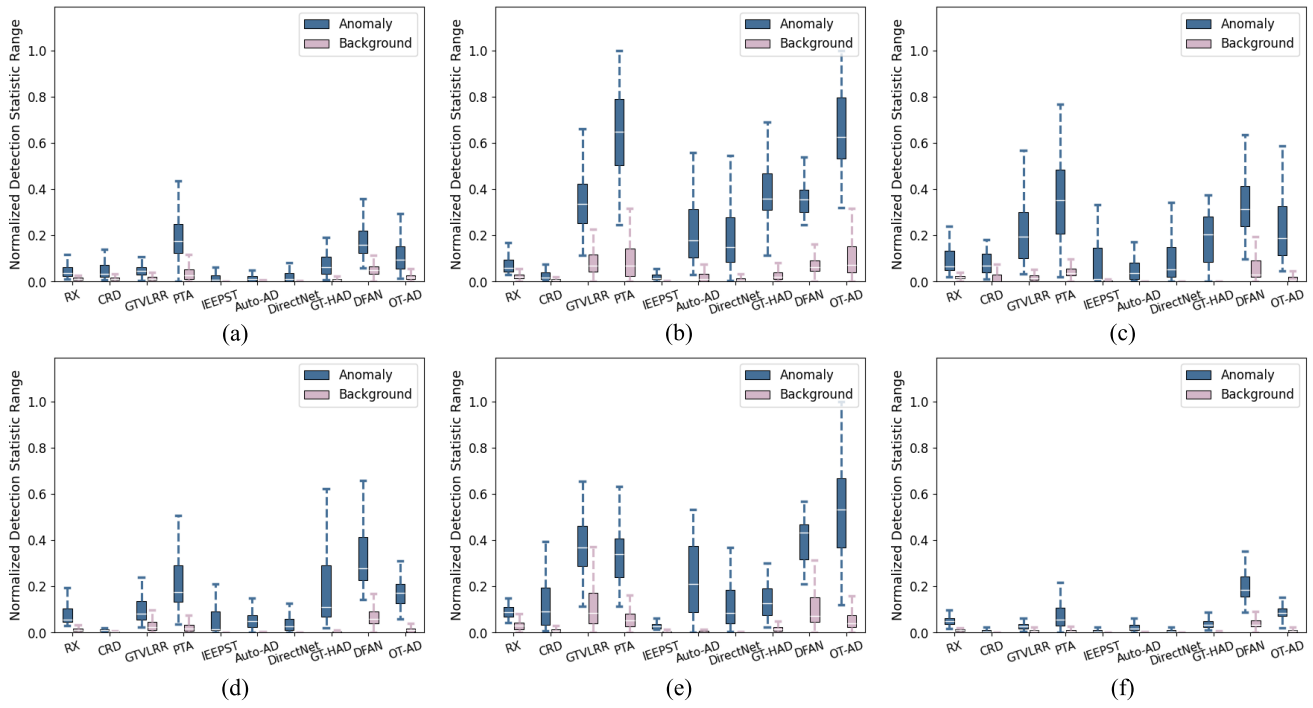


Fig. 8. Box-and-whisker plots of different HAD methods on six publicly available datasets. (a) ABU-Airport-3. (b) ABU-Airport-4. (c) ABU-Beach-4. (d) ABU-Urban-2. (e) ABU-Urban-3. (f) ABU-Urban-4.

Fig. 9 shows the detection maps obtained by the basic network and the network adding the Wasserstein distance constraints derived in Section III-B. It can be seen that there are pieces of background artifacts in the results of the basic network, especially on ABU-Airport-4, ABU-Beach-3, ABU-Urban-3, and HYDICE. Some of the results also contain

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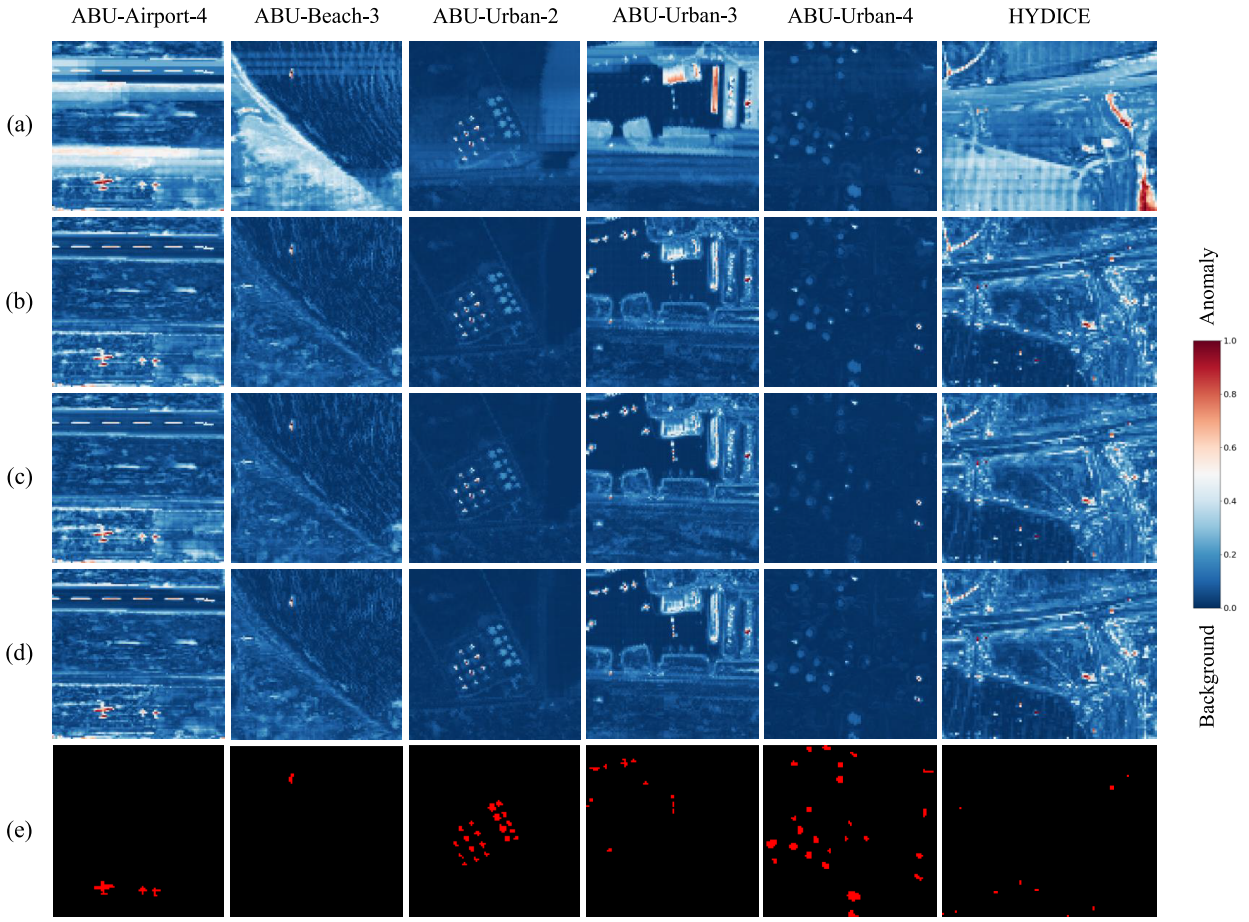


Fig. 9. Anomaly detection results of ablation experiments on six datasets. (a) No Wasserstein distance constraints. (b) Only Wasserstein constraint 1. (c) Only Wasserstein constraint 2. (d) Two Wasserstein constraints exist simultaneously. (e) Ground truth of anomalies.

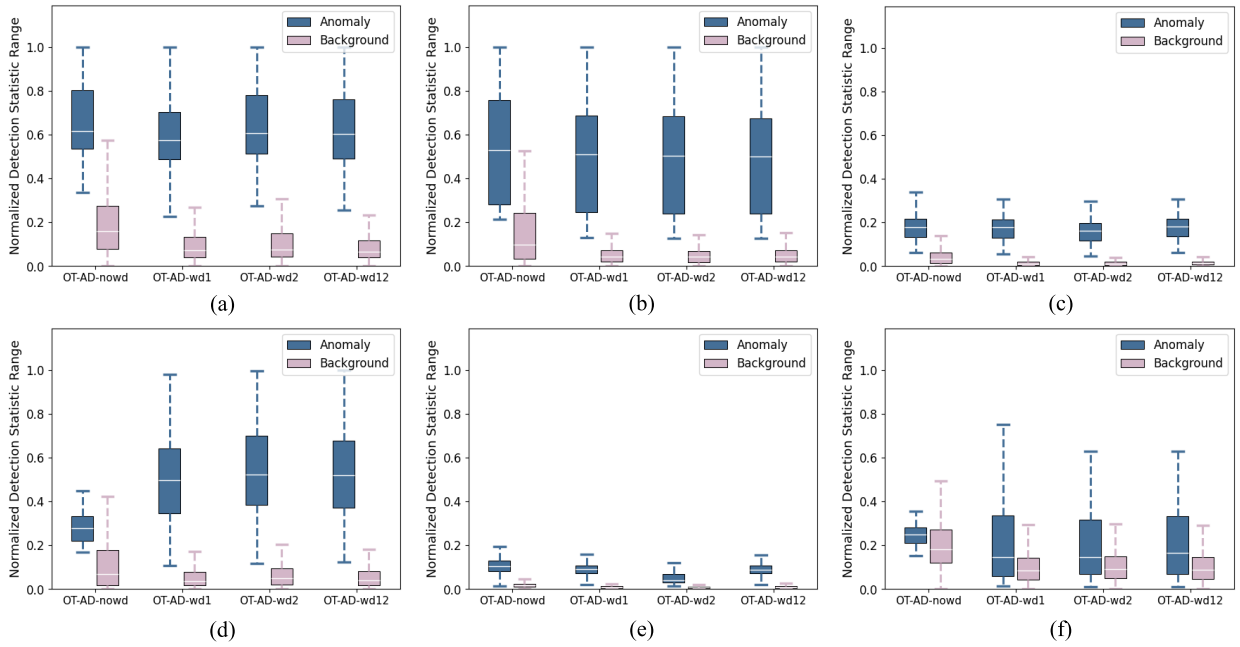


Fig. 10. Box-and-whisker plots under different Wasserstein constraints across the six datasets. (a) ABU-Airport-4. (b) ABU-Beach-3. (c) ABU-Urban-2. (d) ABU-Urban-3. (e) ABU-Urban-4. (f) HYDICE.

edge traces produced during the generation and restoration of image blocks. When we apply the Wasserstein distance to impose constraints on the background, it can be seen that the background is largely suppressed. Fig. 10 is the

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TABLE VIII
RUNNING TIMES (S) OF THE TEN HAD ALGORITHMS ON 16 PUBLICLY AVAILABLE DATASETS

Datasets	RX	CRD	GTVLRR	PTA	IEEPST	Auto-AD	DirectNet	GT-HAD	DFAN	OT-AD
ABU-Airport-1	**0.09**	21.61	149.39	21.23	5.26	*5.45*	4669.79	12.01	17.59	6.53
ABU-Airport-2	**0.09**	21.62	121.13	21.43	*5.69*	5.38	4583.70	11.58	14.16	6.27
ABU-Airport-3	**0.08**	21.42	117.75	24.55	6.21	6.60	4458.12	11.43	14.78	*6.26*
ABU-Airport-4	**0.07**	22.23	142.61	19.98	2.17	*3.47*	5529.78	11.37	14.51	6.27
ABU-Beach-1	**0.21**	49.88	252.56	53.49	*9.14*	7.82	8883.57	29.54	76.52	13.41
ABU-Beach-2	**0.07**	21.16	120.32	22.71	*5.88*	5.31	4498.78	11.22	21.58	6.29
ABU-Beach-3	**0.08**	21.79	125.89	21.68	4.66	*5.93*	3690.44	11.16	11.33	6.10
ABU-Beach-4	**0.06**	44.36	214.65	28.87	5.48	15.64	7744.58	23.95	63.86	*12.59*
ABU-Urban-1	**0.06**	21.36	134.97	20.50	2.62	*5.35*	3891.71	11.12	12.98	6.19
ABU-Urban-2	**0.08**	22.36	117.02	23.98	*4.89*	2.90	5759.36	11.36	17.36	6.33
ABU-Urban-3	**0.08**	21.96	126.23	22.08	5.02	*5.97*	4872.21	11.25	15.12	6.26
ABU-Urban-4	**0.08**	22.42	93.85	24.76	*5.91*	3.04	4028.72	11.19	14.44	6.29
ABU-Urban-5	**0.07**	21.56	125.72	24.42	4.87	7.25	6532.36	11.37	15.26	*6.28*
HYDICE	**0.07**	17.61	98.76	16.38	1.96	13.26	3432.53	7.97	12.20	*4.35*
Cri	**0.22**	319.13	2461.26	121.63	4.73	*40.82*	26714.68	266.498	758.38	75.43
SanDiego	**0.07**	21.17	146.94	18.91	5.32	12.84	5466.39	11.09	16.38	*5.95*
Average	**0.09**	43.23	284.32	30.41	4.99	*9.19*	6547.30	29.01	68.53	11.30

[1] Bold numbers indicate the shortest running time, underlined numbers indicate the second shortest running time, and italic numbers indicate the third shortest running time.
[2] All the results in the table are the average results obtained by these algorithms over five runs, except for DirectNet due to its long training time.

box-and-whisker plots under different constraints. It can be seen that after adding our Wasserstein constraints, the separability of background and anomaly is significantly enhanced, which further illustrates the effectiveness of our method.

E. Model Complexity

This section investigates the algorithmic complexity of all comparison methods. Table VIII shows the running times of the ten methods across different datasets. For OT-AD, we reported the training time required when using the two constraints, with all parameters set to their default values. We can see that RX is the fastest algorithm; this is due to its simple and effective idea, thus becoming a classical algorithm. IEEPST also has a high running speed. GTVLRR and DirectNet run slower. Our OT-AD ranks second among deep learning-based approaches. Tables IX and X compare FLOPs (G) and model parameters (M), respectively. The results show that our OT-AD has relatively low FLOPs and parameters, ranking in the upper-middle among all deep learning-based methods. From the above analysis, we can see that OT-AD achieves a favorable balance between algorithm performance and computational complexity, making it possible for real-time applications.

TABLE IX
FLOPS (G) OF THE FIVE DEEP LEARNING-BASED HAD ALGORITHMS ON 16 PUBLICLY AVAILABLE DATASETS

Datasets	Auto-AD	DirectNet	GT-HAD	DFAN	OT-AD
ABU-Airport-1	11.9698	*0.3581*	**0.0413**	1.7433	0.1378
ABU-Airport-2	11.9698	*0.3581*	**0.0413**	1.7433	0.1378
ABU-Airport-3	11.9698	*0.3581*	**0.0413**	1.7433	0.1378
ABU-Airport-4	11.8174	*0.3465*	**0.0387**	1.5200	0.1306
ABU-Beach-1	26.5196	*0.3440*	**0.0382**	3.3490	0.1290
ABU-Beach-2	11.8392	*0.3481*	**0.0391**	1.5546	0.1316
ABU-Beach-3	11.7848	*0.3440*	**0.0382**	1.4884	0.1290
ABU-Beach-4	24.4143	*0.2724*	**0.0221**	1.0756	0.0844
ABU-Urban-1	11.9589	*0.3573*	**0.0411**	1.7371	0.1373
ABU-Urban-2	11.9915	*0.3598*	**0.0417**	1.7713	0.1389
ABU-Urban-3	11.8174	*0.3465*	**0.0387**	1.5200	0.1306
ABU-Urban-4	11.9698	*0.3581*	**0.0413**	1.7433	0.1378
ABU-Urban-5	11.9698	*0.3581*	**0.0413**	1.7433	0.1378
HYDICE	9.1893	*0.3223*	**0.0333**	0.8973	0.1156
Cri	163.3309	*0.2259*	**0.0117**	1.9370	0.0554
SanDiego	11.7957	*0.3448*	**0.0383**	1.4942	0.1295
Average	22.8943	*0.3376*	**0.0367**	1.6913	0.1251

[1] Bold numbers indicate the least FLOPs, underlined numbers indicate the second least FLOPs, and italic numbers indicate the third least FLOPs.

F. Sensitivity Analysis

We investigated the network performance of the two Wasserstein distance-constrained branches when other parameters were set to their default values, with the weight coefficients α_1 and α_2 ranging within the interval [0, 1] (with an interval of 0.1), and plotted it as a heat map, as shown in Fig. 11. Excluding the zero point, results for 120 sets of parameters

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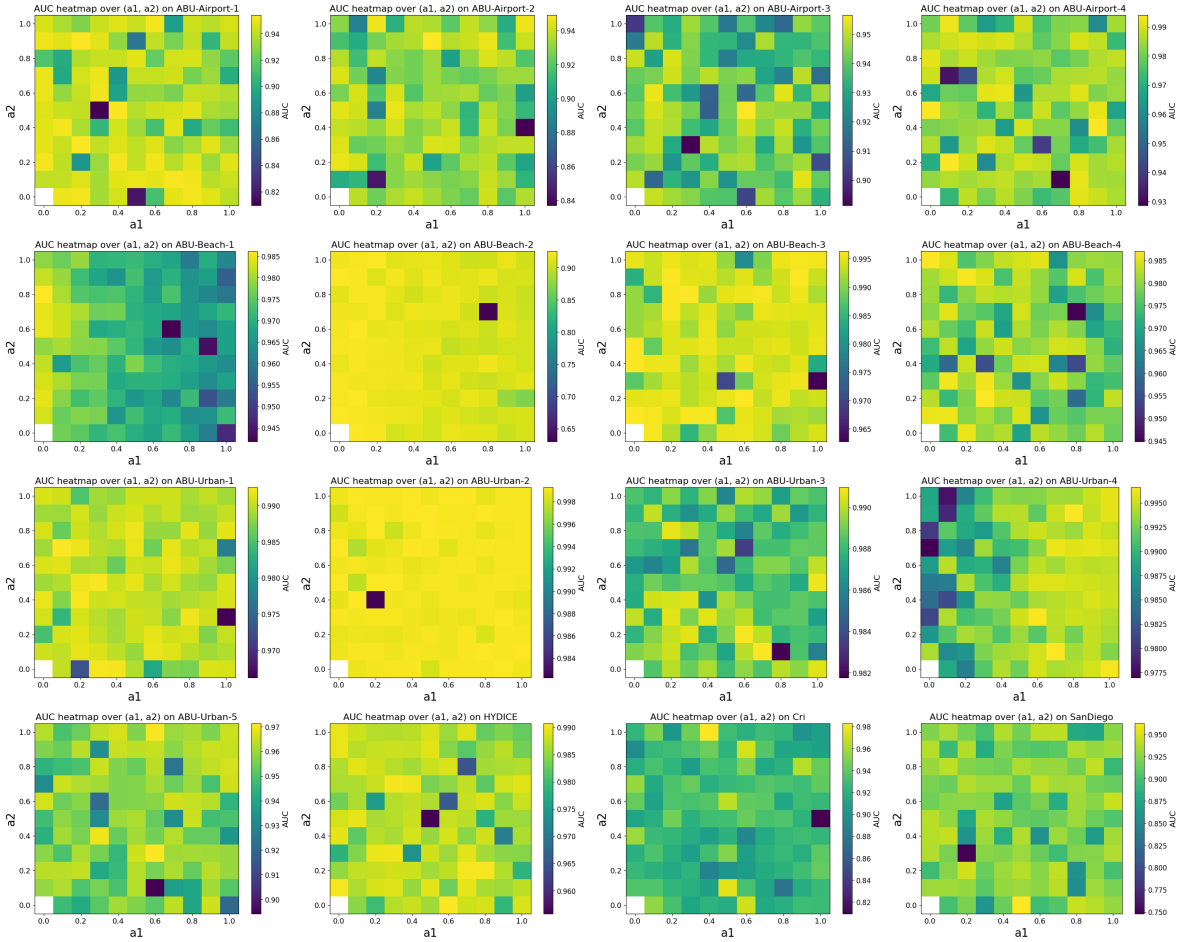


Fig. 11. Sensitivity analysis of weighting parameters (α_1, α_2) on the 16 datasets. The AUC values are plotted at 0.1 intervals for 120 possible (α_1, α_2) values except zero. Brighter colors indicate higher AUC values, while darker colors indicate lower values. The color bars on different datasets have different ranges of values.

TABLE X
PARAMETERS (M) OF THE FIVE DEEP LEARNING-BASED HAD ALGORITHMS ON 16 PUBLICLY AVAILABLE DATASETS

Datasets	Auto-AD	DirectNet	GT-HAD	DFAN	OT-AD
ABU-Airport-1	3.2501	0.4954	0.2553	**0.0901**	*0.3066*
ABU-Airport-2	3.2501	0.4954	0.2553	**0.0901**	*0.3066*
ABU-Airport-3	3.2501	0.4954	0.2553	**0.0901**	*0.3066*
ABU-Airport-4	3.2304	0.4792	0.2392	**0.0790**	*0.2905*
ABU-Beach-1	3.2262	0.4758	0.2357	**0.0775**	*0.2870*
ABU-Beach-2	3.2332	0.4815	0.2415	**0.0807**	*0.2928*
ABU-Beach-3	3.2262	0.4758	0.2357	**0.0775**	*0.2870*
ABU-Beach-4	3.1050	0.3766	0.1366	**0.0274**	*0.1879*
ABU-Urban-1	3.2487	0.4942	0.2542	**0.0898**	*0.3055*
ABU-Urban-2	3.2529	0.4977	0.2576	**0.0915**	*0.3089*
ABU-Urban-3	3.2304	0.4792	0.2392	**0.0790**	*0.2905*
ABU-Urban-4	3.2501	0.4954	0.2553	**0.0901**	*0.3066*
ABU-Urban-5	3.2501	0.4954	0.2553	**0.0901**	*0.3066*
HYDICE	3.1895	0.4458	0.2058	**0.0592**	*0.2571*
Cri	3.0261	0.3120	0.0720	**0.0098**	*0.1233*
SanDiego	3.2276	0.4769	0.2369	**0.0777**	*0.2882*
Average	3.2154	0.4670	0.2269	**0.0750**	*0.2782*

[1] Bold numbers indicate the least params, underlined numbers indicate the second least params, and italic numbers indicate the third one.

are plotted on each small graph. As can be seen, the parameter sensitivity varies across different datasets. For ABU-Beach-2 and ABU-Urban-2, the choice of parameters is not sensitive and presents a uniform distribution throughout the region. ABU-Airport-1, ABU-Beach-3, and ABU-Urban-1 show a slightly uniform parameter performance. For ABU-Urban-4, the choice of parameters tends to the right. For ABU-Beach-1, the choice of parameters tends to the left. Some datasets, such as ABU-Airport-3, are sensitive to parameter values.

G. Failure Case Analysis

As can be seen in Table VI, our method is not a panacea and performs poorly on some datasets, such as ABU-Airport-1 and ABU-Airport-2. We recorded their heatmaps and analyzed them as shown in Fig. 12. Although our method achieves the highest AUC value on ABU-Beach-2, we can see that it still misidentifies many background pixels as anomalies. From the analysis of the data itself, it can be observed that within the red boxes, all these images contain some background objects that appear visually similar to anomalies to the naked eye. Our method mainly imposes constraints on the image distribution in the background reconstruction process, and the most prominent advantage is that it can achieve good

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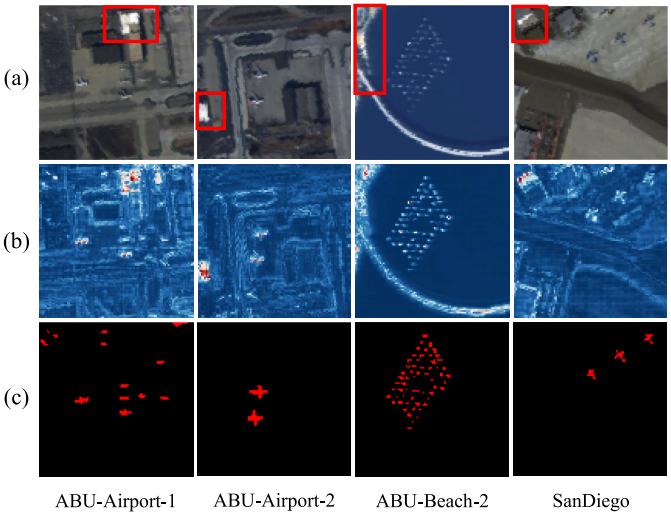


Fig. 12. Analysis of poor detection results. (a) False color images. (b) Detection heatmaps of our OT-AD. (c) Ground truth of anomalies.

detection results on most datasets. However, it may struggle when dealing with some specific approximate phenomena. In the subsequent work, we will analyze and model these special backgrounds, and the scale is also a factor that needs to be considered.

V. Concluding Remarks

In this article, we focus on the HAD task. First, we discuss the statistical distribution characteristics of HSIs. Then, inspired by the mathematical theory of optimal transport, the reconstruction task of the hyperspectral background is modeled from a new angle. The optimal transport of reconstruction and background is converted to which of the reconstruction and original image, and the mathematical proof is given. Combining the proposed theory with the Transformer architecture, we propose a novel OT-AD algorithm. Experiments on a large number of publicly available datasets verify the effectiveness and the superior tradeoff between performance and complexity of our method. In the future, we will continue to explore the characteristics of HSIs and discover more mathematical rules, so as to solve HAD problems more efficiently in more complex scenarios.

References

[1] X. Wei, W. Yu, H. Huang, L. Sun, C. Zhao, and G. Shen, "MAMInet II: Illumination-insensitive modalitywise assimilation guided multistage interaction network for hyperspectral and LiDAR joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539513.

[2] W. Yu, H. Huang, Y. Shen, and G. Shen, "Crossmodal sequential interaction network for hyperspectral and LiDAR data joint classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[3] W. Yu, H. Huang, Y. Shen, L. Sun, C. Zhao, and G. Shen, "IamCSC: Intuitive assimilation modality driven crossmodal subspace clustering for land-cover identification and hyperspectral-LiDAR fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5505013.

[4] X. Wei, W. Yu, R. Liu, M. Ma, C. Zhao, and G. Shen, "HugIpuNet: Hybrid updating graph motivated illumination-wise property unification network for hyperspectral and DSM joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5522314.

[5] H. Zhang, X. Han, J. Deng, and W. Sun, "How to evaluate and remove the weakened bands in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502515.

[6] X. He, C. Tang, X. Liu, W. E. Zhang, K. Sun, and J. Xu, "Object detection in hyperspectral image via unified spectral–spatial feature aggregation," *IEEE Trans. Geosci. Remote Sens.*, 2023, Art. no. 5521213.

[7] F. Jiang et al., "Adaptive center-focused hybrid attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503916.

[8] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag. (replaces Newsletter)*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[9] T. Goudemant et al., "Onboard anomaly detection for marine environmental protection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7918–7931, 2024.

[10] S. Kumari, S. Agarwal, N. K. Agrawal, A. Agarwal, and M. C. Garg, "A comprehensive review of remote sensing technologies for improved geological disaster management," *Geological J.*, vol. 60, no. 1, pp. 223–235, Jan. 2025.

[11] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[12] E. Bedini, "The use of hyperspectral remote sensing for mineral exploration: A review," *J. Hyperspectral Remote Sens.*, vol. 7, no. 4, pp. 189–211, Dec. 2017.

[13] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[14] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[15] H. Sun, L. Wang, L. Zhang, and L. Gao, "Hyperbolic space-based autoencoder for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522115.

[16] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[17] K. An, "Sulla determinazione empirica di una legge didistribuzione," *Giorn Dell'inst Ital Degli Att*, vol. 4, pp. 89–91, Feb. 1933.

[18] N. Smirnov, "Table for estimating the goodness of fit of empirical distributions," *Ann. Math. Statist.*, vol. 19, no. 2, pp. 279–281, Jun. 1948.

[19] M. B. Wilk and R. Gnanadesikan, "Probability plotting methods for the analysis of data," *Biometrika*, vol. 55, no. 1, pp. 1–17, Mar. 1968.

[20] G. Monge, "Mémoire sur la théorie Des déblais et Des remblais," *Mem. Math. Phys. Acad. Royale Sci.*, vol. 1781, pp. 666–704, 1781.

[21] L. V. Kantorovich, "On the translocation of masses," *Dokl. Akad. Nauk. USSR (NS)*, vol. 37, pp. 199–201, 1942.

[22] E. F. Montesuma, F. M. N. Mboula, and A. Souloumiac, "Recent advances in optimal transport for machine learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 2, pp. 1161–1180, Feb. 2025.

[23] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, vol. 30, 2017, pp. 5998–6008.

[24] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. ECCV*, 2020, pp. 213–229.

[25] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021, pp. 611–631.

[26] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6877–6886.

[27] H. Kwon, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, p. 3342, 2003.

[28] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[29] H. Su, Z. Wu, A. Zhu, and Q. Du, "Low rank and collaborative representation for hyperspectral anomaly detection via robust dictionary construction," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 195–211, Nov. 2020.

[30] S. Wang, X. Wang, Y. Zhong, and L. Zhang, "Hyperspectral anomaly detection via locally enhanced low-rank prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 6995–7009, Oct. 2020.

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[31] Y. Xu, B. Du, L. Zhang, and S. Chang, "A low-rank and sparse matrix decomposition- based dictionary reconstruction and anomaly extraction framework for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 7, pp. 1248–1252, Jul. 2020.

[32] X. Geng, K. Sun, L. Ji, and Y. Zhao, "A high-order statistical tensor based algorithm for anomaly detection in hyperspectral imagery," *Sci. Rep.*, vol. 4, no. 1, p. 6869, Nov. 2014.

[33] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[34] X. Yang, B. Tu, Q. Li, J. Li, and A. Plaza, "Graph evolution-based vertex extraction for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17372–17386, Dec. 2024.

[35] X. Shen, H. Liu, J. Nie, and X. Zhou, "Matrix factorization with framelet and saliency priors for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504413.

[36] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505016.

[37] T. Wu et al., "Spectra-difference based anomaly-detection for infrared hyperspectral dim-moving-point-target detection," *Infr. Phys. & Technol.*, vol. 128, Jan. 2023, Art. no. 104489.

[38] B. Tu, X. Yang, W. He, J. Li, and A. Plaza, "Hyperspectral anomaly detection using reconstruction fusion of quaternion frequency domain analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 8358–8372, Jun. 2024.

[39] E. Bati, A. Çalışkan, A. Koz, and A. A. Alatan, "Hyperspectral anomaly detection method based on auto-encoder," *Proc. SPIE*, vol. 9643, pp. 220–226, Oct. 2015.

[40] C. Zhao, X. Li, and H. Zhu, "Hyperspectral anomaly detection based on stacked denoising autoencoders," *J. Appl. Remote Sens.*, vol. 11, no. 4, Sep. 2017, Art. no. 042605.

[41] K. Zhao and L. Zhang, "Spectral–spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 92, pp. 166–176, Aug. 2018.

[42] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Mar. 2020.

[43] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[44] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.

[45] L. Wang, X. Wang, A. Vizziello, and P. Gamba, "RSAAE: Residual self-attention-based autoencoder for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510614.

[46] P. Xiang, S. Ali, J. Zhang, S. Ki Jung, and H. Zhou, "Pixel-associated autoencoder for hyperspectral anomaly detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103816.

[47] Y. Huo, X. Cheng, S. Lin, M. Zhang, and H. Wang, "Memory-augmented autoencoder with adaptive reconstruction and sample attribution mining for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518118.

[48] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[49] S. Lin, M. Zhang, X. Cheng, L. Shi, P. Gamba, and H. Wang, "Dynamic low-rank and sparse priors constrained deep autoencoders for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–18, 2024.

[50] X. Cheng et al., "Deep feature aggregation network for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

[51] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, S. Hu, and X. Liu, "Robust multi-stage progressive autoencoder for hyperspectral anomaly detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 134, Nov. 2024, Art. no. 104200.

[52] W. Xie, J. Lei, B. Liu, Y. Li, and X. Jia, "Spectral constraint adversarial autoencoders approach to feature representation in hyperspectral anomaly detection," *Neural Netw.*, vol. 119, pp. 222–234, Nov. 2019.

[53] W. Xie, B. Liu, Y. Li, J. Lei, C.-I. Chang, and G. He, "Spectral adversarial feature learning for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2352–2365, Apr. 2020.

[54] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[55] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[56] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[57] D. Wang, L. Gao, Y. Qu, X. Sun, and W. Liao, "Frequency-to-spectrum mapping GAN for semisupervised hyperspectral anomaly detection," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1258–1273, 2023.

[58] Z. Wang, X. Wang, K. Tan, B. Han, J. Ding, and Z. Liu, "Hyperspectral anomaly detection based on variational background inference and generative adversarial network," *Pattern Recognit.*, vol. 143, Nov. 2023, Art. no. 109795.

[59] J. Wang, S. Guo, Z. Hua, R. Huang, J. Hu, and M. Gong, "CL-CaGAN: Capsule differential adversarial continual learning for cross-domain hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517315.

[60] N. Huyan, X. Zhang, D. Quan, J. Chanussot, and L. Jiao, "AUD-Net: A unified deep detector for multiple hyperspectral image anomaly detection via relation and few-shot learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 5, pp. 6835–6849, May 2024.

[61] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3631–3645, Feb. 2025.

[62] S.-S. Young, C.-H. Lin, and Z.-C. Leng, "Unsupervised abundance matrix reconstruction transformer-guided fractional attention mechanism for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9150–9164, May 2025.

[63] J. Ma, W. Xie, and Y. Li, "Exploring hyperspectral anomaly detection with human vision: A small target aware detector," 2024, *arXiv:2401.01093*.

[64] Z. Wu and B. Wang, "Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508015.

[65] X. He et al., "Global-to-local spatial–spectral awareness transformer network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530422.

[66] K. Yu et al., "Accelerating hyperspectral anomaly detection with enhanced multivariate gaussianization based on FPGA," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535512.

[67] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[68] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[69] X. Sun, L. Zhuang, L. Gao, H. Gao, X. Sun, and B. Zhang, "Information entropy estimation based on point-set topology for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5523415.

[70] D. Wang, L. Zhuang, L. Gao, X. Sun, X. Zhao, and A. Plaza, "Sliding dual-window-inspired reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504115.

[71] R. Mcgill, J. W. Tukey, and W. A. Larsen, "Variations of box plots," *Amer. Statistician*, vol. 32, no. 1, pp. 12–16, Feb. 1978.



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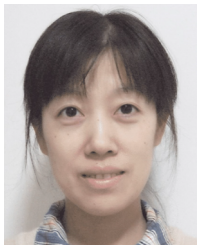
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